VU AI HUB · LECTURE NOTES

Modelling Human Behaviour

Year 1, February to March 2026

Written by Inaaya and shared on Notion. Original page: cheddar-wall-f3b.notion.site/2fa9801a930980ee9e9ce616aacc705d

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3rd of February

Models and Human Behavior L1

AI is fundamentally about behavior modelling - language models mimic how we talk, recommenders predict what we want, robots mimic how we learn. All AI systems are based on either theories or data of how humans behave.

Understanding human behavior gives us power to:

- predict outcomes
- design better systems
- influence positive change
- notice when others exploit it

Algorithms are tools for modelling humans

Apps are platforms for using models to influence used behavior

AI systems are testbeds for our ideas of how humans think and act - because of the failure and successes of AI is how cognitive science existed

Modelling human behavior is useful for:

- the future of AI: growing awareness of the limitations of LLMs
- addressing societal challenges: understanding what causes undesired behavioral patterns and how to address them
- technoethics: assessing the impact on human behavior

Models

A model is a simplified representation of a system, entity or phenomenon

- they're informative or useful for a specific purpose

Types of models

- Physical models: tangible, 3D objects you can physically interact with
  - Strengths - directly manipulable, communicated ideas intuitively

- Computational models: represent real or hypothetical systems/entities and their dynamics
  - Strengths - simulate complexity and change over time, explore many scenarios cheaply
- Mathematical models: use equations and formulas to represent relationships between factors
  - Strengths - precise, testable
- Conceptual models: organize ideas into frameworks or categories
  - Strengths - organize thinking, aids systematic implementation of goals

Different scales, different rules

- Micro scale - individual person
- Meso scale - group or organizational level (ex. group of people)
- Macro scale - system or aggregate level (ex. cultural trends)

Choice depends on what you want to explain, the categories are fuzzy and context-dependent

*Often can't just integrate explanations across scales

Risks of using models

A model is a simplification

- During the design of a model, certain choices were made
  - assumptions
  - limitations
- A model is not reality

Models help us to:

- Explain, predict, communicate, experiment, classify and strategize & intervene

It does this by:

- Specifying the conditions for achieving the outcomes we want
- Identifying relationships between variables
- Representing structures and connections
- Quantifying differences between categories
- Showing what happens (if our theories are accurate)

Theory, data and models

Theory: A broad, well-supported explanation of how phenomena are related

Model: A concrete manifestation of a theory, something we can use to solve a specific problem

Hypothesis: What we expect to see if we run a simulation or implement a model

Data: What actually happens

Approaches to making models

Theory-first (deductive)

Theory identifies relationship between variables (ex. variable rewards release dopamine)

Model implements theory into something concrete (ex. designing app notifications)

Hypothesis defines expectations from theory (ex. fake notifications will significantly increase user engagement in a month)

Test against data, refine, iterate

Data-first (deductive)

Collect data, identify patterns

Define correlations as relationships (ex. people click faster on the color green)

Capture relationships in model (ex. make important buttons green)

Test model on real-world data

(develop theory)

Data-driven (inductive) approach

With large datasets, you might build predictive models directly from data (like neural networks)

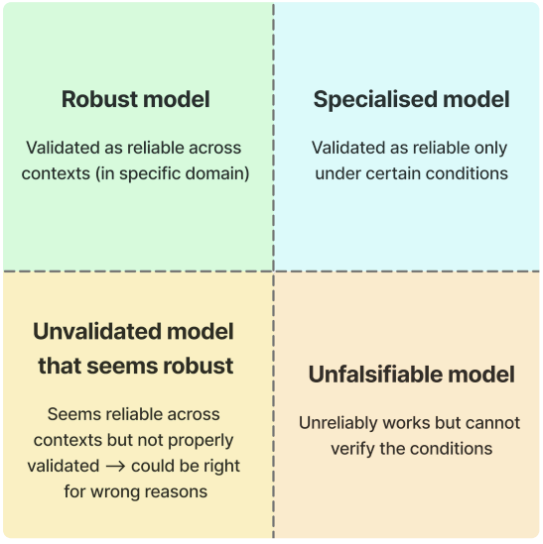
Can then work to understand what hypotheses that model has implicitly learned

Risk of overfitting: may not capture essential properties

Building useful models

If all models are wrong, how do you at least ensure your model is useful?

1. it generates testable hypotheses
   - *if model is correct, what can we expect?*
   - *if the model is wrong, how can we tell?*
2. validating them on data
   - *are the outcomes as expected?*
   - *does it account for all phenomena?*
3. see whether the model can generalize
   - *does it work equally well on new data?*
   - *does it work under different conditions?*
4. be transparent about limitations
   - *BMI doesn't count for muscle mass, body composition, ethnicity*
   - *LLM statistically models human language use, but does not model human cognition per se*
5. use with caution
   - *weather forecasting: accuracy decreases beyond 7-10 days*
   - *relying on financial growth predictions*



⬆️ = reliable ⬇️ = unreliable ⬅️ = generalizes ➡️ = doesn't generalize

5th of February

Models and Human Behavior L2

Theoretical Approaches to Behavior

Why do we do things?

- many dimensions from personal to situational factors, physical to psychological

Cognitive vs Behavioral Approaches

- Explaining behavior in terms of observable and latent factors

Behaviorist

Focuses strictly on observable relationships between external stimuli and behavior (if X changes, Y happens), how the one influences the other. It relies on mindless associations and habits rather than conscious thought. There are some factors that can be directly measured by an external observer:

- Independent variables
  - the presence/absence of a physical stimuli (ex. a colour, sound, object etc.)
  - a specific environmental setup (ex. layout of elements in a space etc.)
- Dependent variables
  - a quantifiable change in behavior (ex. doing the thing or not, having the time etc.)

Influencing the approach

- If X has a predictable impact on Y, manipulate X to get the desired outcome
- Can first create a new relationship between X and Y through repeated exposure to cause + effect
- Theory/model examples: classical conditioning and operant conditioning

Applications of behaviorist models

To influence human behavior: marketing, interface design

To mimic human behavior: RL algorithms

Cognitive

Makes inferences about unobserved mental states (thoughts, feelings, beliefs) and how they drive behavior, it focuses on the subjective experience and conscious choice. The factors are the same as with behaviorist models but including:

- an affective experience (ex. fear, anger)
- a psychological construct (ex. confidence, motivation, personality)
- a conscious thought process that leads to a decision (ex. I believe that doing X will make Y happen)

All of these could either be independent (ex. how changes in confidence affects someone's behavior) or dependent (ex. how a puppy affects someone's depression)

Cognitive explanations

How unobserved mental factors (+ observed factors) influence observed behavior

Influencing the approach

- Change thought processes (ex. changing their beliefs with evidence or logic) to achieve a behavioral outcome
- Use measurable stimuli (ex. give compliments) to influence affective states or other psychological constructs (confidence), and thereby change behavior

- Theory/model examples: motivational (ex. self-determination theory), decision-making (ex. rational choice theory)

Applications of cognitive models

To influence human behavior: therapy chatbot, digital wellbeing apps

To mimic human behavior: rules and statistical approximations of "human-like" reasoning mechanisms

Behaviorist vs. Cognitive

Behaviorism assumes some mental states (ex. enjoying a reward or a desire to look like a celebrity), just fails to explain them

Key differences:

- Focus: behaviorist approaches only focus on observable outcomes, cognitive also on underlying mechanisms
- Strategy: harnessing subconscious impulses, habits, or associations with stimuli (behaviorist) vs. convincing or persuading someone (cognitive)

These distinctions are useful for thinking about technoethics (ex. does taking more steps lead to better health), as well as the foundational limitations of AI (ex. do computers actually think/feel, or just seem that way)

Classifying variables

Observable factors: directly measurable physical stimuli or environmental conditions (speed, weight, temperature etc.)

Latent factors: subjective psychological states or constructs requiring inference or self-report (emotions, beliefs, personality etc.)

- Can't measure directly, indirectly measured or inferred using observables like self-report scales

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Social Influence

Humans are social beings, evolved to care about the actions and judgments of others. These social-relational factors affect not only people's momentary feelings and actions, but also their overall performance and wellbeing.

Social behaviors are not only about the presence of others, but relational: the unique dynamic between people, their feelings towards/assumptions about each other, the nature of their relationship. An important phenomenon in models involving social influence is emergence: patterns in group behaviors that can't be predicted from individual behaviors or sentiments alone.

Our behavior is not only determined by our own mental state or the behavior of others, but also what we assume about the mental states of others. These assumptions can be based on many things: their behaviors, the cultural context, your own past experiences, beliefs etc.

The media equation: people instinctively respond to media as if it were real life, even when they know better (ex. getting scared from horror). A special case of this is applying social expectations to technology, treating computers as if they were sentient beings.

Anthropomorphism: attributing humanlike qualities to non-human things, and increasingly so the more convincingly 'alike' and humanlike they seem. This can be used as a form of behavior manipulation, with social robots and chatbots for example, it makes many people trust them more and respond more politely to it.

Social-relational factors

Properties of social context, relationship, and disposition towards another, some examples:

Observable:

- Number of friends/connections
- Frequency of interactions
- Proximity
- Duration of marriage

Latent:

- Perceived closeness
- Intimidation
- Respect or admiration
- Quality of relationship

10th of February

Agent-Based Modelling L1

It is a form of computational modelling whereby a phenomenon is modelled in terms of autonomous agents, their local interactions, and their environment, allowing population-level patterns to emerge from individual behavior.

(Aggregate models on the other hand represent a population as a single quantity that changes over time, they often ignore the individual differences (heterogeneity), local interactions and spatial structure)

- An important model is the population
  - Why? Population size directly affects demand for food, water, energy, housing healthcare and education
  - Long-term population dynamics influence economic development, urbanization, something
  - Models help us test different scenarios
- In 1798, Thomas Robert Malthus proposed an exponential growth model for living group with enough resources (human, or animal)
  - P(t)= P(0)* e^rt
    - Population at time = initial population * population growth rate
    - It is tricky: heterogeneity, local interactions and emergence
    - Aggregate models: they abstract away heterogeneity

In agent-based modelling, a phenomenon is represented using the following core components:

- An agent is an autonomous individual or object with particular properties, actions and possibly goals
- Environment is the structured context in which agents are situated and with which they interact
- Interactions, refers to local agent- agent and agent-environment processes trough which information is exchanged and behavior is updated
- And optionally links, links are explicit relational connections between agents that structure who interacts with whom

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Fundamental concepts within ABM context:

Emergence

It is the big picture that appears when many agents interact, even if no single agent intended for it to happen

- Like when you have a swam of birds, we can see the whole swarm on macro-level, but the birds itself only see up to 10 other birds or so and don't see the whole

Emergence in Schelling's Model

- A model proposed by Nobel laureate Thomas Schelling to investigate how individual actions shape global outcomes

1. A population of agents belonging to different social groups are placed in a spatial environment.
2. Each agent has a simple local preference:
    - It is happy if a certain fraction B of its neighbors belong to agent's group, otherwise it's unhappy
3. If an agent is unhappy it moves to an empty location, otherwise it stays.
4. The model runs until most agents are happy

Segregation outcome

On average, agents have about 90% of the same-type neighbors

Feedback loops

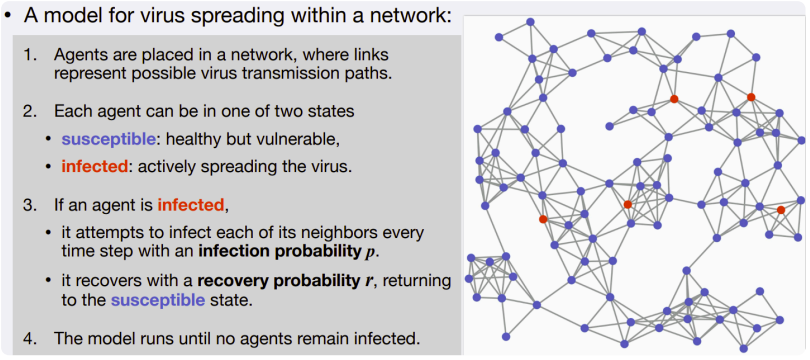
An agent's behavior is influenced by the behavior of others, which then changes the environment for everyone else

- ex. In an evacuation, an agent sees "visibly panicked neighbors" and becomes afraid, which might cause them to run, making even more neighbors panic

Non-linearity

Non-linearity means that small changes in inputs or parameters can lead to disproportionately large or qualitatively different system outcomes in the final result.

In Schelling's model *B* is the minimum fraction of similar neighbors needed for an agent to be happy, and because of non-linearity the actual percentage is higher than *B*



Path dependence

Path dependence means that the final state of the model is determined by the specific sequence of events and interactions that happened during the simulation, meaning that two simulations with the same parameters but different starting positions may end in different final states.

Modeling opinion dynamics

A model to understand how repeated interactions shape the spread and clustering of opinions over time. Agents only interact and average their opinions if they are already sufficiently similar (within a "confidence bound").

1. Agents hold an opinion represented by a number between 1 and 0 (two extremes)
2. Each agent has a confidence bound that limits which other opinions it considers
3. Agents interact and average their opinions, but only if they are sufficiently similar

4. The model runs until opinions no longer change

17th of February

Networks L1

Networks can be seen as an approach to study complex systems where individual micro interactions shape macro level patterns. It defines networks as structures existing above the individual level but belonging to the population level, allowing us to represent relationships between components.

- A complex system is one where interactions between components shape outcomes, so studying components separately is insufficient

Network Representation and Analysis

Networks (ex. Gmail) are formally defined by vertices (nodes (ex. email addresses)) and edges (links (ex. emails)), They can be represented in several ways:

- Visualization: graphic drawings useful for intuition but prone to distortion at scale.
- Adjacency matrix: a mathematical grid indicating connections, fast for checking links but memory intensive.
- Adjacency list/ Edge list: efficient for small networks where most nodes aren't connected.

Approaches for analyzing networks

- Exploratory: quantifying and comparing properties like node connectivity (often contextualizing them with a random baseline)
- Explanatory: using statistical models to examine why certain ties exist
- Predictive modelling: fitting models to predict future edges or missing attributes
- Causal modelling: identifying cause-effect relationships through mechanistic simulations or interventions

Degree distribution

The degree of a node is the number of edges connected to it. Many real world networks exhibit heavy tailed distribution, where few hubs have many connections and most of them have very few

Preferential attachment: a rich get richer mechanism where new nodes are more likely to connect to existing nodes that already have high degrees

- Barabasi Albert Model: A specific model demonstrating how this mechanism leads to emergence of hubs.
  1. The network starts with a small connected seed of *m0* nodes
  2. At each time step, one new node is added to the network
  3. The new node creates m links to existing nodes
  4. Each link connects to node i with probability proportional to its degree *ki*

Paths and the small world phenomenon

Paths and distances capture how close or far apart nodes are in a network. A path is a sequence of nodes such that each node is connected to the next node along the path by an edge

Geodesic path: the shortest distance between two nodes

Diameter: longest shortest path (or length)

Average path length $\langle d \rangle$: the average of the shortest paths length between all pairs of nodes

Six degrees of separation

The claim that any two people can be connected through only a handful of intermediate contacts → Milgram Experiment

Watts-Strogatz model: explains the small world phenomenon, how a regular lattice can achieve very short average path lengths by adding just a few random shortcuts

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Transitivity and Triadic closure

Transitivity refers to the presence of triangles in a network. If A knows B and B knows C, does A and C know each other? (Is the friend of my friend also my friend?)

Clustering coefficient: measures the density of these triangles at both global and local scales

Triadic closure: the social principle that people with a common friend are likely to become friends themselves. This is driven by opportunity (shared meetings) , trust, and social incentives (reducing the stress of disconnected friend groups)

19th of February

Networks L2

Node Centrality

A large volume of research on network has been devoted to the concept of centrality: which are the most important (or central) nodes in a network?

Degree

The node degree can be used as a centrality measure, capturing immediate connectivity of a node and reflects its direct access to others in the network

- It is a local type of centrality: it depends only on the node's neighborhood topology and does not account for paths beyond one hop
- In social networks, degree can be interpreted as measuring a person's opportunities to interact, access information, or exchange resources with others

Closeness

Closeness centrality measures how close a node is to all other nodes in the network

- It is a global type of centrality: it depends on the distances between a node and every other node
- In social networks, closeness reflects how quickly a person can interact and reach others in the network

Betweenness

Betweenness centrality measures the extent to which a node lies on shortest paths between pairs of other nodes in the network

- It is a global type of centrality: it depends on shortest paths between all pairs of nodes
- In social networks, betweenness is often interpreted as capturing brokerage potential

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Meso-scale structures in networks

We want a simplified model of something very complicated (networks are often too large and complex to be analyzed without some kind of simplification). Real-world networks are heterogeneous (different in kind), many use:

- Extrapolation (make prediction about unseen nodes)
- Interpolation (identify missing links)
- Generalization (nodes of this type are like the others of the same type)
- Mechanisms (how did this network arise? what are the underlying mechanisms?)
- Explanations (coarse-graining: a modeling technique that simplifies complex, high-resolution systems by grouping atoms into fewer interaction sites)
- Useful division (finding groups to assign treatments)
- Simplifications (regression models need ranks or groups)

Communities

A group of nodes that have a higher likelihood of connecting to each other than to nodes from other communities, we can partition a network in different ways:

Hypotheses of community detection

H1) Fundamental hypothesis: A network's community structure is uniquely encoded in its wiring diagram

H2) Connectedness and density hypothesis: A community is a locally dense connected subgraph in a network

H3) Random hypothesis: Randomly wired networks lack an inherent community structure

H4) Maximal modularity hypothesis: For a given network the partition with maximum modularity corresponds to the optimal community structure

- Intuition: modularity checks if nodes inside the same community are connected more often than chance would predict
- Implication: a high modularity indicates dense connections inside communities and relatively few connections between communities

Community detection algorithms

- Clauset-Newman-Moore: greedy modularity
  - General idea: start with every node being in its own group, then recursively and greedily choose a pair of groups to merge based on modularity (the pair that would produce the greatest increase or smallest decrease in the modularity)
    - Advantage: simple to implement and understand
    - Disadvantage: it can be a bit slow
- The Louvain Method
  - Start with every node being its own group

    Phase 1:

    1. For each node i:
       - Evaluate the gain in modularity if node i joins the community of its neighbors j
       - The node i is then placed in the community of maximum gain

    Phase 2:

    - Create a network from the communities found in Phase 1
    - Use this new network as the input for the Phase 1 until there are no more changes

24th of February

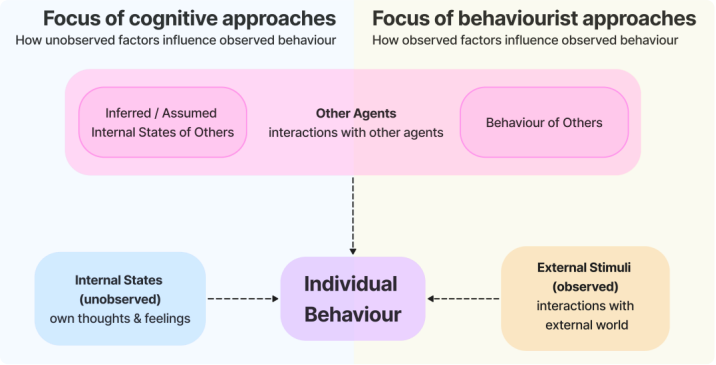
Influencing Behavior with UI Design Patterns L1

Preventing errors

Influencing people's behavior is not just about persuasion or manipulation, but about ensuring the basic functionality of a system. The primary causes of errors that were occurring was not technical malfunction but 'human error', people doing the wrong things.

Systems were way less user-friendly than they are now, users were expected to become experts themselves. When there is one user error it's the user's fault, but when there are many errors systematically that's when the designer is probably at fault and in need of a better model.

Human factors engineering: The focus moved from writing manuals and labels to understanding how to better engineer every single touch, button, interaction and create a good symbiosis between humans and machines.



Many theories/models of human behavior ignored embodiment (how your body enables you to think, feel, experience and interact with the world) because we assumed that humans were neutral information processors: that we perceive the world as is and act logically based on this information.

- However, we do not have a perfect memory, do not see the world objectively, and do not process information deterministically like computers

Understanding human cognitive limitations

- We use heuristics (mental shortcuts) not because we're lazy, but because we must
  - An expert is someone who learns the right heuristics to solve a problem fast
- The working memory holds around 5-9 information items at a time (George Miller)

Dual Systems Theory

We are capable of deep thinking, but it's not our baseline

- System 1 vs System 2 thinking
  - System 1 = intuitive, automatic responses that take minimal effort (being on autopilot)
  - System 2 = effortful focus and reasoning, deliberate action
- Humans use System 1 way more than System 2, due to the principle of economy (we evolved to save on mental resources)

3rd of March

Data-driven models

5th of March

The Ethics & Implications of Modelling People

Accuracy is difficult to establish when it comes to modelling human behavior, because of how complex humans are. Accuracy is also not all that matters, the more effective a model is at predicting and influencing human behavior, the greater the risk of manipulation.

Risks of Human Behavior Modelling

HBM's are normative, not just descriptive

- Modelling involves choices: what to include, what to ignore and how to measure it, we ask normative questions (how things ought to be), not just descriptive ones (how things are).
  - A student engagement model that flags low participation in class discussions, taking it as a lack of interest
    - Implicit: that extroverted, oral communication styles are the norm for serious students
  - Many concepts used in human behavior models (ex. intelligence, trustworthiness, productivity) are themselves not objective properties but social constructs, they originated in particular historical and institutional contexts
    - They contain assumptions about what we value in human life that are contested
- When a model treats constructs like intelligence as if they were measurable natural properties, it can make a normative position seem like an objective, scientific one
- We tend to associate quantification with objectivity: not because numbers are more accurate, but because they appear impersonal and scientific
  - hides subjective/normative choice in what to measure and how
- The appearance of objectivity only makes assumptions harder to see and thus harder to contest

HBM's can be self-fulfilling

- People are affected by the categories applied to them
  - As social creatures, how we perceive ourselves is affected by how others see/treat us, which affects how we behave
- When a model classifies a person as high-risk, low-ability etc., they enter a social world that is organized around that classification
  - Institutions treat them differently, other people treat them differently etc.
- As a result, they may come to understand and present themselves differently
  - School ability streaming: students placed in low streams may receive lower quality instruction, lower teacher expectations, and lower access to advanced material
    - Initial scores could be impacted by many unrelated situational factors
- Internalized oppression: people from marginalized groups unknowingly adopt the negative stereotypes, beliefs, and biases of the dominant culture about their own group, reinforcing the very systems that oppress them

- Stereotyping effect: when people are aware of a negative stereotype about their group's performance, the fear of confirming it impairs performance
- Awareness of the model produces the outcome it predicted (e.g. added anxiety and cognitive load) not through any fixed property of the person
- Recommender systems use behavioral trace data to predict the kinds of content people like

When influencing humans, the ends do not justify the means

- If a behavior change model is very effective, should we use it?
- Engineering of consent: explicit program of using psychological models to shape mass behavior in directions chosen by corporations and governments
- How you treat people (the means) matters at least as much as what you get them to do (the ends)
  - Dehumanizing to manipulate and exploit a person's vulnerabilities
  - People have the right to autonomous choice and informed consent

HBM's are instruments of power, not just knowledge

- Models of human behavior are never created in a social vacuum, they inherit and amplify existing power relations
  - Purpose of a model is defined by those who have the power to use it
  - Validated using data of a world already shaped by power imbalances
  - The people who bear the greatest cost from a model's errors are usually those who don't have the power to negotiate or change it — may not even be aware of how they are modelled
- Algorithmic biases: algorithms learn from biased data, reinforce stereotypes, makes harder to challenge them

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The future of human behavior modelling

As our methods advance, models of human behavior increase in resolution:

- Modelling general patterns in human behavioral dispositions → what affects a specific individual's behavior across situations
- Amplifies existing risks, poses new ones

This allows for mode targeted predictions and interventions

- From ads and design patterns that harness general psychological vulnerabilities, to targeting specific personality types, to advertising specific content to a person at a specific time of day/month when they seem more likely to be in a certain emotional state

Beyond behavioral traces on specific platforms, move towards greater cross-platform synchronization → integrating behavioral trace data across a user's digital ecosystem.

- Agentic AI (systems that takes real-world actions on a person's behalf) requires this integration to function

Risks of wearable data

Wearable devices extend the behavioral trace into the body and physiological states, they can track:

- Heart rate variability →
- Patterns in sleep quality/stages →
- Menstrual cycles →
- Accelerometer data →
- Even phone camera can analyse micro-changes in skin colour (blood flow) to infer changes in physiological and affective states

Internet of Things

Smart home devices generate a continuous record of domestic behavior that is qualitatively different from anything a platform can observe

- (ex. when you wake or go to bed, if you're home alone or not, what/when you eat, whether your routine is changing etc.)
- Valuable data for life/home insurance models: offer discounts in exchange for smart home monitoring , reward desirable patterns

Dialogue agent conversations

Ongoing dialogues with LLM-based agents keeps a record how a person thinks, reasons, expresses emotion, and uses language in a way that is unprecedented in the history of human behavior modelling, they even have the potential to make diagnoses.

- They don't just pay attention to what people say, but more how they say it
  - Sentence length variability, semantic coherence across turns, use of absolute language, first-person singular pronouns, and temporal orientation
    - All of these carry diagnostic signal → can be linked to depression, anxiety, bipolar disorder, early cognitive decline, personality structure, big life events, etc.
- Over time, a conversational AI system holds a model of a person's psychological life that is more detailed and accurate than anything previously available outside a clinical relationship
  - Unlike a clinical relationship, it is not governed by professional ethics, confidentiality obligations or the consent frameworks that protects patients
  - User may not even be aware or knowingly consent to a continuous psychiatric assessment

Combining digital trace data

- Taken together: move towards a dynamic, unified, cross-context individual behavior model that integrates their digital behavior, physiological state, domestic routines, and inner psychological life
- Individual behavior model may become the most valuable and the most dangerous artefact in the contemporary data economy

When too good becomes bad:

- Effectively influencing user behavior → undermines autonomy and other life-goals/interests
- Good for one person → bad if everyone does same thing at once
- AI mimics humans effectively → deepfakes, over-trust, cognitive atrophy
- Accurate model of current world state → amplifies bias, deters systemic change
- Accurate models of a person's desires/weaknesses→ addiction, compulsive shopping, emotional manipulation

10th of March

Review & Wrap-Up

NetLogo en Networkx doornemen

Week 1

Models are simplified representations of reality, informative for a specific purpose

Differrent types of models (physicial etc.) and scalews of analysis (micro, macro, meso)

Different appraches to builind models ans basic criteria for building reliable models

Theoretical approaches to explaining behavior, the effects of psychological and environmental factors as well as the special effect of other people

Week 2

Agent based models simulate population-level patterns by defininf simple behavioral rules for individual agents and telling those interactions play out

We used several exmaples to show how this modelling apprach

Some fundamental concept

complex social outcomes can arise from simple behaviroral

Week 3

12th of March (Guest Lecture)

Beyond the metrics

Attention to audiences would lead to:

- "Dumbing down" of news
  - Making it easier to understand, using less complex wording
- Loss of journalistic autonomy
  - Journalists are listening to audiences more instead of only believing in their own expertise

Prior journalists would have some idea in mind and write with that in mind, they would have an instinct, but don't know for sure. Now there are numbers available which makes it easier for journalists, because they can see how many people look at their news and understand what they're talking about. → Known as rationalizing of audience understanding.

Dominant assumption: clicks are indication of users' interest in news, the metrics showed that users were more interested in junk news (celebrities) than public affairs (politics). By studying these metrics they filmed interviews where participants think-aloud while navigating on their phone. Clicking on news did show that there was some kind of interest, but not necessarily deep interest, sometimes you click because you have to know what the title means but you're not really interested. Not clicking on news did not indicate a lack of interest, sometimes people don't need to click on certain news articles anymore because they've seen it so often and they do not feel the need to read the same thing again, this doesn't mean that they find it less interesting or important. People can also just browse on the news without actually clicking on every single article. There is also a social function of the news, which means that people click on certain news to be informed when talking to others.

To what extent is 'time spent' a good proxy to how engaged people are?

1. It doesn't reflect how the time is spent (tabs can be open without actively looking at them)

- Looking at the 'time engaged' is also not a perfect measure, because people can be engaged but not actively

2. Less time spent can point to being experienced or skilled in using their phone
   - They can save articles quickly and read it later but the actual time spent is brief because they can do it fast
   - They can scan information efficiently without spending a lot of time

Metrics are still valuable, but they are designed to measure and monetize behavior, not understanding user experiences

Capturing perceives worthwhileness and emotional experience

- Both positive and negative emotions

Final take-aways

- Metrics indispensable for understanding news behavior
- Not just a neutral instrument → metrics affects subsequent behavior
- Flatten the human experience around news consumption
- Critically assess them by (holistically) capturing the behavior in everyday context